OLMo2 1B degeneration progression (50 prompts)

2026-04-05 01:37 UTC

Stage Map

- **Base**: `allenai/OLMo-2-0425-1B`
- **SFT**: `allenai/OLMo-2-0425-1B-SFT`
- **RLVR1**: `allenai/OLMo-2-0425-1B-RLVR1`
- **Instruct**: `allenai/OLMo-2-0425-1B-Instruct`

Shared Contract

This progression bundle compares multiple checkpoints under one rollout-statistics contract. The stage summaries were validated to share temperature, number of generations, max token budget, max model length, dtype, trust-remote-code flag, loop detector, and tracked statistics before plotting the progression.

Headline Read

Average loop fraction moves from 40.8% on Base to 5.3% on Instruct.

Average max-length-hit fraction moves from 61.9% on Base to 7.6% on Instruct.

MATH-500: loop fraction moves from 37.8% (Base) to 2.4% (Instruct); max-length-hit fraction moves from 42.2% to 3.0%.

AIME 2024/2025: loop fraction moves from 44.8% (Base) to 5.4% (Instruct); max-length-hit fraction moves from 48.0% to 7.2%.

GPQA Diamond: loop fraction moves from 51.2% (Base) to 4.2% (Instruct); max-length-hit fraction moves from 61.6% to 8.8%.

MMLU-Pro: loop fraction moves from 47.0% (Base) to 14.0% (Instruct); max-length-hit fraction moves from 62.6% to 18.6%.

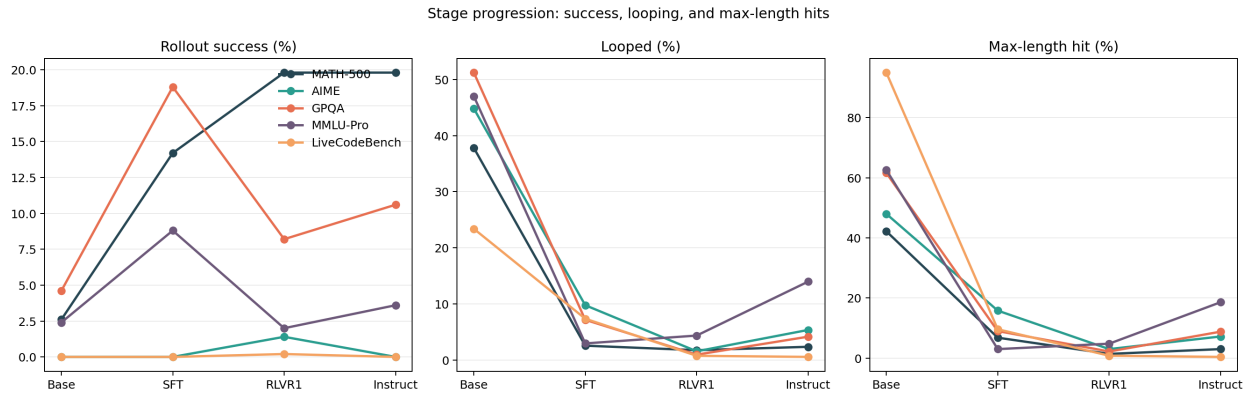

Figure 1: Progression of rollout success, loop rate, and max-length-hit rate across stages, broken out by dataset.

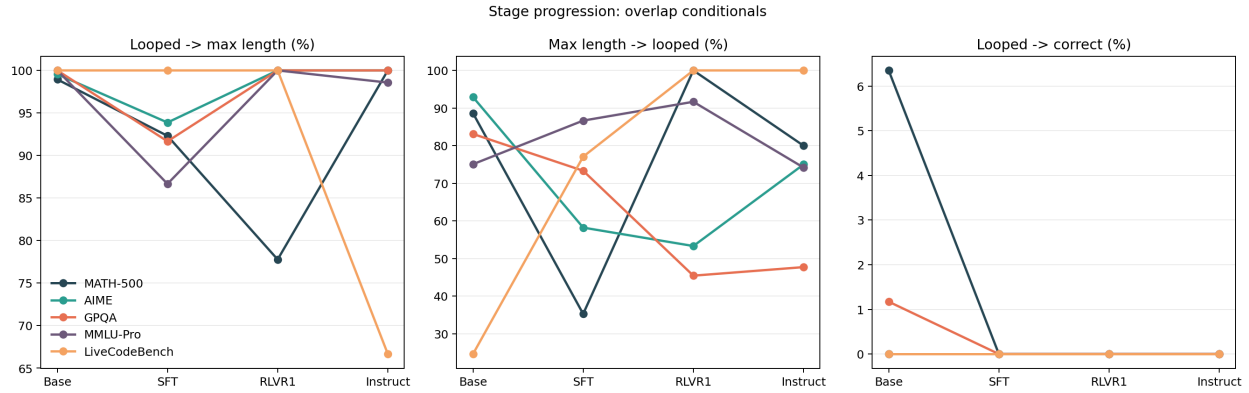

Figure 2: Progression of overlap conditionals across stages, broken out by dataset.

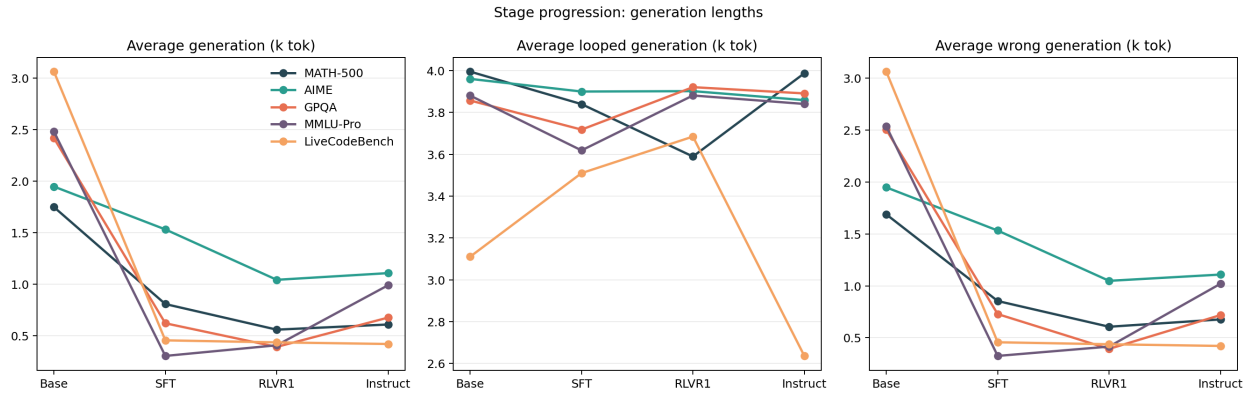

Figure 3: Progression of generation-length statistics across stages, broken out by dataset.

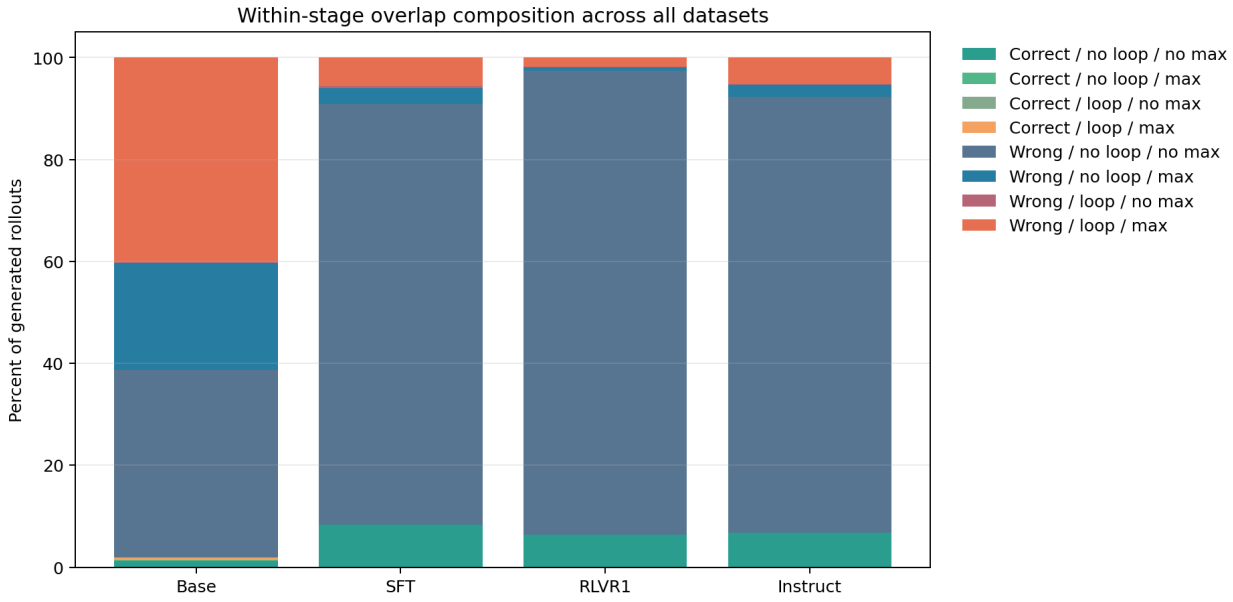

Figure 4: Within-stage overlap composition aggregated across all datasets. This uses the saved raw counts to decompose generated rollouts by correctness, loop status, and max-length-hit status.

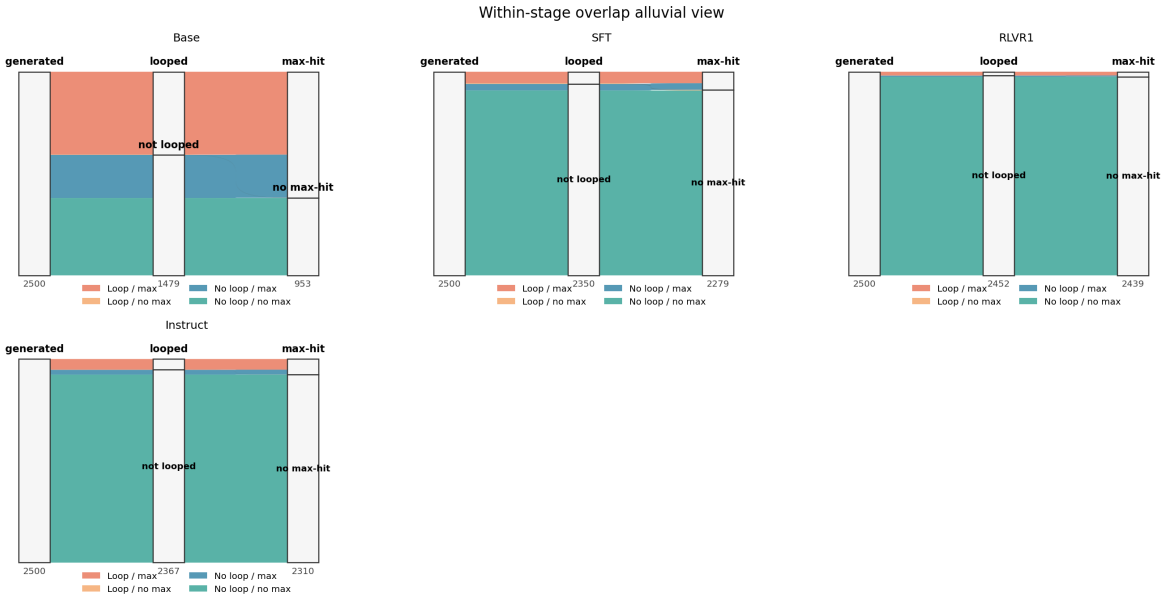

Figure 5: Within-stage alluvial view of the overlap structure. Each stage shows how generated rollouts split into looped versus non-looped branches and then into max-length-hit versus non-max-length-hit branches.

Interpretation

The progression figures answer the exact object of the note: where the degenerate-rollout regime enters the model family under a fixed rollout contract. The line plots are the stage comparison surface; the overlap-composition panel makes the within-stage structure visible without collapsing everything to one scalar rate. If the loop and max-length-hit curves are near-zero in the base stage and then rise sharply after SFT or RLVR, that is direct evidence for a stage-linked introduction of degenerate rollouts. If they are already substantial in the base model, the hypothesis weakens.